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Introduction to Bluetooth

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Connecting PalmPowered handhelds using Bluetooth™

Functional overview, software solutions, and development tools for the Palm OS® Platform.



Agenda Slide

- Overview
- Usage Scenarios & Demos
- Architecture
- Components
- Developer Tools



Bluetooth Quick Facts

Data Transmisson	1 Mbs
Actual throughput	< 720 Kbs
Radio band	2.4 GHz
Distance	30 feet
Line of sight	Not required
Frequency hopping	1600 hops/sec
Certified products	250 +
Bluetooth SIG members	2000 +

The Bluetooth Market Today

- Ramp-up slower than expected, happening now
- Signs of take-off for 2002
  - 10-13 Million units shipped in 2001
  - Positive attitude returning in Press
  - $5 price point achieved by several manufacturers
- Key products with integrated Bluetooth are shipping
  - Cell phones, printers, laptops, network access points

	2001	2002 (est.)
Bluetooth chipsets	13.4 M	40 M

* Source: Cahners InStat, Dec '01



Expect Bluetooth and 802.11

- 802.11
  - Pros: Speed, enterprise adoption
  - Cons: Power, size
- Bluetooth
  - Pros: Mobility, flexibility, cost
  - Cons: Lower speed, newer technology
- Growing industry agreement for co-existence
- Regional differences



Bluetooth (Also) Fits Palm OS Vision

- Traditional handheld criteria
  - Simple, expandable, wearable
  - Long battery life
- Growing need: connectivity while mobile
  - Access information and critical data any time
  - In office, at home, on the road
  - No cable hassles
- Bluetooth connects complementary devices
  - “It takes two to tango”
  - Enhances traditional functionality

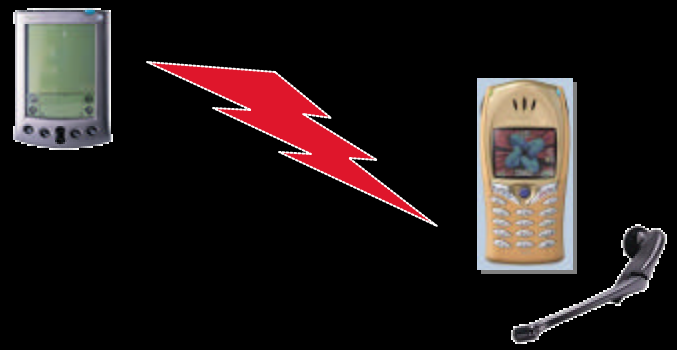


Today's Top 10 Functions

- Exchange data with other users
- Synchronize your data to a laptop
- Connect to the Internet via cell phone
- Send and receive e-mail
- Access data via corporate network
- Play multi-player games
- Print easily
- SMS, Instant messaging
- Dial phone numbers
- Run collaborative applications

Usage Scenarios and Demos

Cell phone connections



Collaboration



Network / Internet access



General interoperability



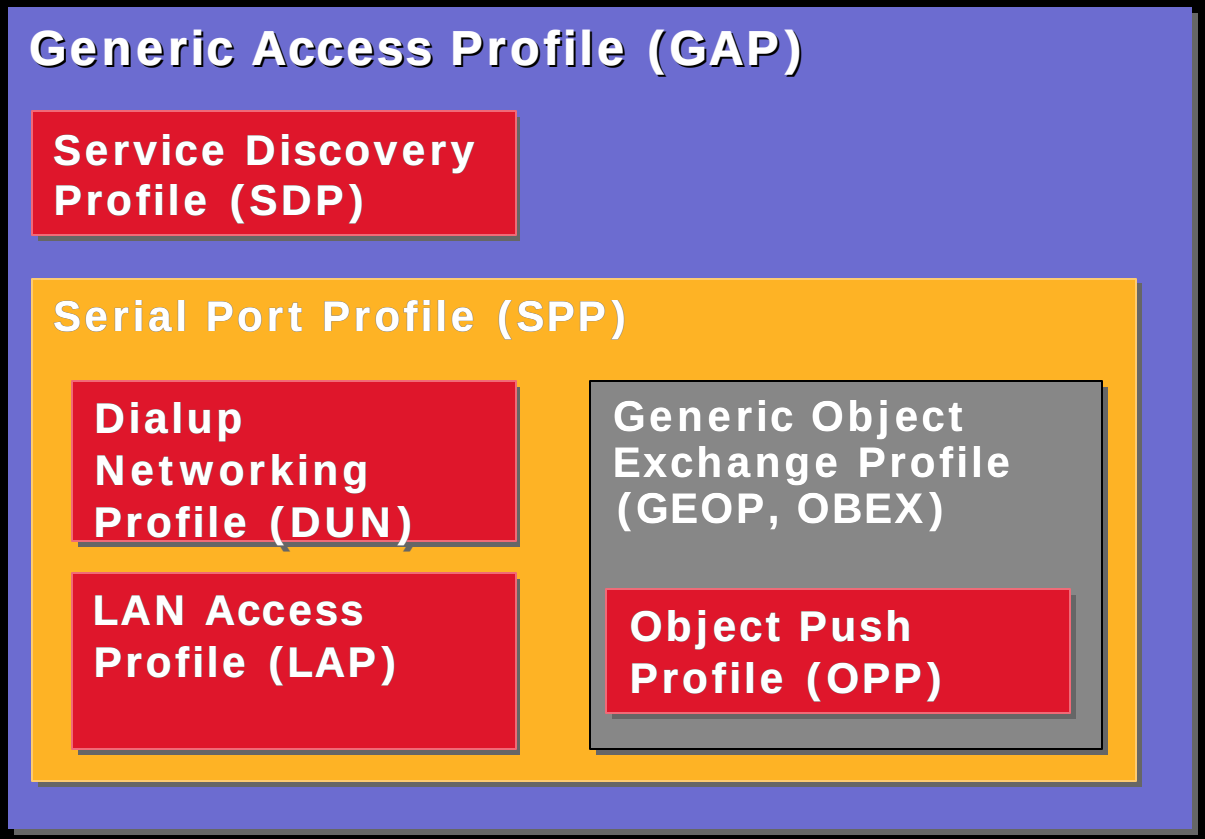


Product Description

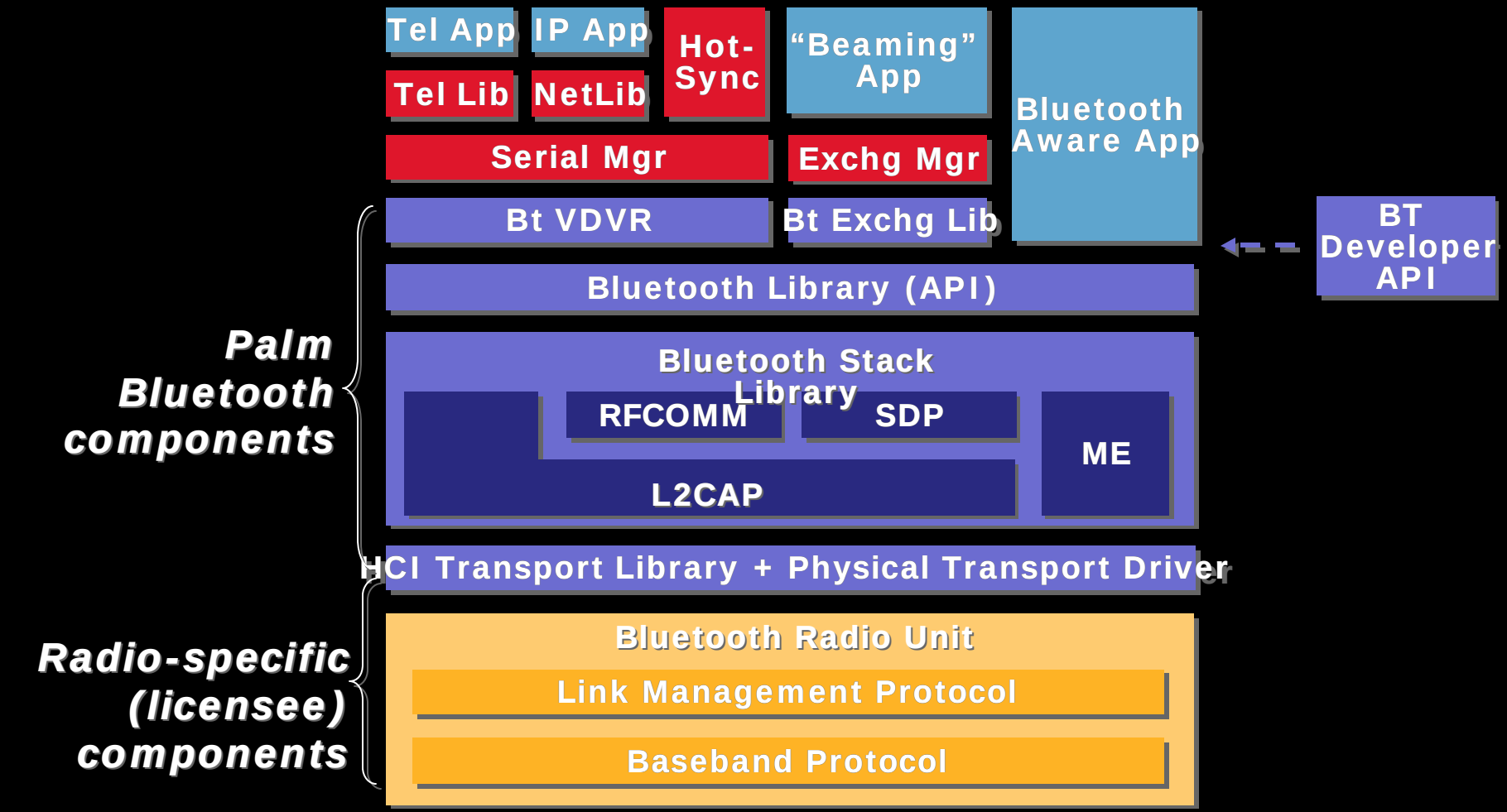
- Certified Bluetooth™ 1.1 stack
- Add-on software for Palm OS 4.1
- May be installed in device ROM or installed in RAM via HotSync® operation
- Available in six languages (English, French, Italian, German, Spanish, Japanese)
- Provides serial and cell phone connections, HotSync®, Internet/network access
- Supports data exchange, collaborative apps, games
- Handles device discovery, configuration, pairing, passkey entry, sending data, and more

Palm Bluetooth Profiles

How functionality is standardized



Bluetooth Architecture





Palm Bluetooth Components

- Virtual Serial Driver (VDRV)
- Exchange Library (“Send” menus)
- Bluetooth and Connection panels
- Sample applications
  - Piconet and Games
- Bluetooth stack and upper-level API
- Transport drivers



Toolkits

- Standard SDK
  - 3 ways to write applications
    - Exchange Manager, VDRV, Palm Bluetooth API
    - Technical documentation
  - Complete stack, samples, tools
  - Available from www.palmos.com/bluetooth
- Compatible HDKs from major suppliers
  - Red-M, TDK Systems, Northstar, Palm Inc.
  - All kits are usable with POSE



Potential Applications

- Bluetooth is an enabler, not the solution
- Next generation Personal Area Network applications
  - Enterprise, education, health care, leisure
  - Collaboration, Multi-player games
  - Automotive, hands-free telephony
  - Data streaming, audio, images
  - Secured transactions, payment terminals, m-commerce
  - Remote control, device management
- Wanted: Killer Apps
  - Palm Bluetooth now on PalmOS.com
  - Developer kits available now



Conclusion

- Expect more connectivity than ever before
  - Smartphones and 2-piece solutions
  - Ad-hoc peer-to-peer connections
  - 802.11 and Bluetooth for access at different times
- Future versions
  - ARM version of Palm Bluetooth
  - Enhancements and evolution
- Bluetooth at PalmSource
  - Technical session, Labs
  - Hardware and software development kits available

Win a Bluetooth Developer Kit

Red-M Blade Palm OS® developer kit give-away

- 100 free kits for PalmSource delegates to win
- Visit the Red-M booth #116 to enter
  - Completed forms to be returned at Red- M booth
- Drawing will take place on Thursday 7th
  - Location – Developer Court, Civic Auditorium
  - Time - Lunch Break 1:00 pm



Making it easier to develop Bluetooth applications for Palm OS

red-M

Bluetooth™



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